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Sustainability through materials: A review of green options in construction

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ABSTRACT

In light of the growing worldwide consciousness regarding environmental issues and the depletion of resources, attaining sustainability in the construction sector has become a necessity. This study carefully examines how environmentally friendly materials support sustainable building practices, emphasizing three key areas of investigation: the broad spectrum of feasible green materials, the quantitative differences in emissions between green and conventional materials, and the overall effect of these materials on the economy and health. The study employs a robust quantitative approach, leveraging numerical optimization and detailed analyses to explore a wide array of environmental resources, including recycled, renewable, and low-impact options. A comparison between materials is completed, offering refined insights into the environmental benefits of green materials across the entire life cycle-from manufacturing to lifecycle. The primary findings highlight the observable benefits of using these materials and demonstrate how they can drastically lower emissions and environmental effects. It offers verifiable proof of the viability of removing dangerous exposures and promoting healthier indoor environments by utilizing environmentally friendly materials. The graphical information and numerical findings that are provided in this paper form the basis for putting workable and significant strategies into action. The results support the broad use of green materials by providing evidence of their beneficial effects on economic growth, environmental preservation, and human health. The strong quantitative evidence highlights the applicability of these suggestions and offers a convincing case for incorporating sustainable practices in the building industry.

1. Introduction

Human activities are causing serious environmental damage, including tsunamis, wildfires, floods, and droughts. These are all exacerbated by global warming, which is causing the sea level to rise and the ozone layer to deplete. Depleted ozone increases the risk of skin cancer, and can have an adverse effect on the plantation and soil. Greenhouse gases also impact the ozone layer. For instance, carbon dioxide can contribute to the development of polar stratospheric clouds, which can quicken the rate of ozone depletion (Arnesson, 2019). A source of carbon dioxide production is the building construction sector. The construction industry is booming in urban areas due to the growing number of people living in cities (Ghalandari et al., 2023). In 1950, there were 751 million urban dwellers; by 2030, that number is expected to reach 5 billion leading to a dramatic rise in construction, and hence construction materials. Millions of tons of wastage are generated annually from these building supplies. Greenhouse gases are exacerbated throughout the construction process, from raw material extraction to final disposal (Cheng et al., 2016). The materials used in a building's envelope (outer shell of the building), can have a significant impact on the amount of energy the building uses and the amount of greenhouse gases it emits. The materials selected, how they are used, and the quality of construction all have a part in determining a building's energy performance (Agencia Internacional de Energía, 2018). Emissions from the construction industry are largely attributable to the manufacturing of carbon-intensive materials like concrete and steel (Batayneh et al., 2007). According to research conducted in 2010 by Hui Yan (Hui Yan et al., 2010), the production, transport, and use of energy for construction equipment account for 98.6–99.2 percent of all emissions. Of this total, 81.6–86.7 percent are embodied GHG emissions, 6.1–8.4 percent are from transportation for building materials, and 6.4–8.6 percent are from energy use. Mavi et al. (Kiani Mavi et al., 2021) claim that construction has a disproportionately harmful impact on the environment, and this problem is more pronounced in less developed nations. In Malaysia, construction accounts for 24% of CO_2 emissions (Klufallah et al., 2016), in India, it accounts for 53.4 % of national CO_2

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emissions (Sharma et al., 2011), and in Nigeria, emissions from construction and manufacturing industries increased from 2557 to 23714 Gg of CO_2 equivalent between 2000 and 2015, an approximately 827% increase (FRN, 2018). According to the research conducted by Tirth et al. (2019), steel, bricks, cement, and concrete collectively account for roughly 66% of the aggregate emissions, while other construction-related sources account for 74% of all emissions. According to a 2014 study by Guzman et al. (Solís-Guzmán et al., 2009), construction accounts for 40% of annual waste production and materials consumption in the European Union. Based on the findings of the 4th Assessment Report conducted by the Intergovernmental Panel on Climate Change (IPCC) (Reay et al., 2007), it was determined that greenhouse gas (GHG) emissions originating from buildings accounted for a total of 8.6 billion metric tons of carbon dioxide equivalent (t-CO_2-e) in the year 2004. According to Mardiana et al. (SB, 2015), projections indicate that the total carbon dioxide equivalent emissions (t-CO_2-e) could potentially reach a value of 15.6 billion by the year 2030. This would result in a 26% rise in carbon dioxide emissions, which would account for around 30–40% of the overall greenhouse gas emissions. In a study conducted by Ezema et al. it was found that buildings contribute to approximately 33% of global carbon dioxide (CO_2) emissions associated with energy use, amounting to 7.85 gigatons (Gt). It is anticipated that the emissions will experience a rise from 11 to 15.6 Gt by 2030. (Ezema et al., 2016). The materials used in external walls, upper floor constructions, and ceilings are particularly emissions-intensive, accounting for 84.2% of the total emissions from a building. For example, the embodied energy of steel is about 32 MJ per kilogram and for cement is about 7.8 MJ per kilogram. If the consumption of construction materials does not change, the world's cement production could reach 3.5 billion metric tons by 2050. Fig. 1 depicts the CO_2 emission condition of the major construction materials, as found in studies of Wu et al. and Zhu et al. (Wu et al., 2023) (Zhu and Chen, 2010) regarding the matter of environmental pollutants and life cycle energy consumption associated with the manufacturing of materials utilized in residential building. It is evident that aluminum produces larger quantity of carbon-dioxide emissions and timber produces higher rate of waste.

Over time, scholars have directed their attention towards the integration of building materials that are reusable, recyclable, or biodegradable. This approach aims to mitigate the negative impacts of waste generation, pollution, and energy consumption. Examples of such materials include wood, earthen materials, bricks, crushed or recycled concrete, Gypsum or Drywall, Straw Bale Insulation, Wool Carpet, and Linoleum Flooring. The study conducted by Irshidat and Al-Nuaimi examines the effects of using carbon dust, a byproduct of aluminum facilities, into the production of cementitious composites (Irshidat and Al-Nuaimi). Work on the physical characteristics of a three-component mineral binder, consisting of hydrated lime and cement bypass dust, which is a residual material generated during cement manufacturing has been done (Owsiak et al., 2020). Glass debris was utilized as a fine aggregate by Jing et al. (2020) in architectural mortar, while Wang et al. (2020) examined the micro-properties and mechanical properties of strain-hardening cementitious composites incorporating recycled brick powder. The study conducted by Srimahachota et al. (2020) presents compelling evidence on the considerable potential of utilizing recycled nylon fibers derived from discarded fishing nets as a reinforcing agent in cement-based products. The utilization of plastic waste in various manners is elucidated by webb et al. (Webb et al., 2013), who conducted a study on the influence of plastic biomineralization on the resilience of plastic-reinforced brickwork. Wang et al. investigated the interfacial bonding activity between steel and sawdust by analyzing the mechanical and micro characteristics of cementitious composites (Liu et al., 2021). Multiple studies have been conducted by researchers worldwide to investigate carbon emissions resulting from the construction of different types of structures. A compilation of select studies is presented in Table 1 and the LCA method is also shown in Fig. 2. Table 2 displays a compilation of research that have been examined in relation to the carbon footprints of different construction materials.

In order to reduce the environmental ramifications of the construction sector, it is imperative to foster the advancement of novel construction materials characterized by reduced emissions. The sustainable development strategy pertaining to construction materials involves the efficient utilization of those resources, while simultaneously dealing with the adverse environmental consequences associated with their production and utilization (Frontczak and Wargocki, 2011). A green building material is designed for the same purpose and eventually the green building movement seeks to design, construct, and operate buildings that have a minimal impact on the environment and human health (Fithian and Sheets). Guna et al. (kumar Guna et al., 2021) explained how to make gypsum ceiling tiles using wool and coconut coir, two inexpensive and biodegradable resources. Sugarcane fiber and oil palm empty fruit bunch (OPEFB) fiber have both been claimed to be useful as wall insulation against the cold (Rama Rao and Ramakrishna, 2022). Several studies have shown that bio-based insulating concretes can be made by recycling agricultural lignocellulosic materials and natural fiber as bio-aggregates (Asdrubali et al., 2015)– (Sui and Liu,

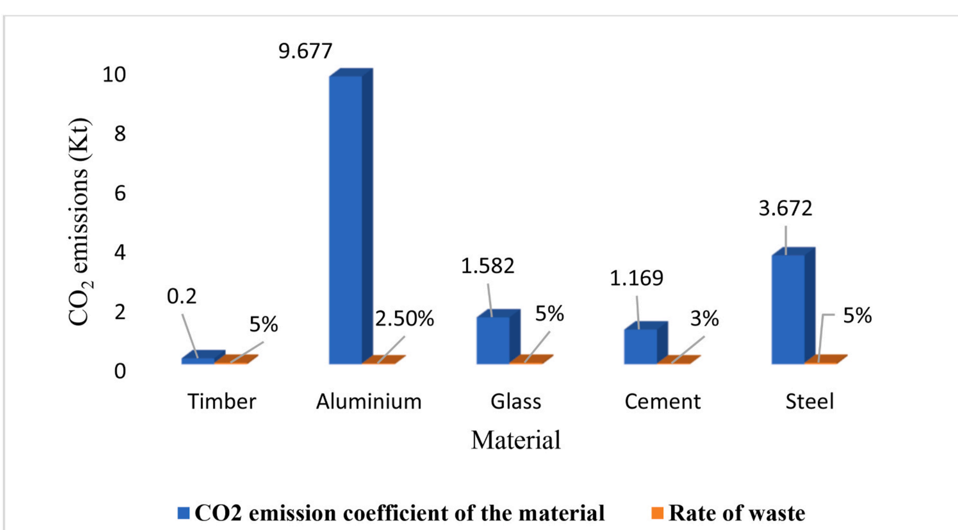

Fig. 1. Construction waste rate and carbon dioxide emission coefficient.

Table 1
Analyzed data from a range of construction projects.

Type of Structure	Main material used for construction	The method used for assessment	Quantity of CO_2	country	Reference
Residential Building	Reinforced concrete structure	LCA Process	8707004 kg CO_2e	China	Hong et al. (2015)
Single-story training center	Masonry-wall Timber frame wall	LCA Process	432 kg CO_2/m^2 363 kg CO_2/m^2	UK	Fu et al. (2014)
Residential building	Masonry Steel	LCA process	189 kg CO_2e/m^2 164 kg CO_2e/m^2	Belgian	MATHUR et al. (2021)
Residential Building		LCA process	430 kg CO_2e/m^2 (average value)	Germany	HolgerC. L (2012)
Houses	Mud brick Cement blocks	ICE database	228 kg CO_2/m^2 397 kg CO_2/m^2	Cameroon	Henry et al. (2014)
Residential Building	Concrete, wood	LCA Process	400 kg CO_2e/m^2	Sweden	Brunklaus et al. (2010)
Residential Building	Reinforced concrete frame	Survey and LCA Process	2395 kg/m^2	Nigeria	Ezema et al. (2016)
Detached dwelling Multi-occupancy dwelling		LCA Cradle to occupation	264 kg CO_2/m^2 360 kg CO_2/m^2		Nässén et al. (2007)
Residential Buildings	Bricks based Not mentioned	LCA process	246 kg CO_2e/m^2 257 kg CO_2e/m^2		Ortiz et al. (2010)
Lightweight timber frame Heavyweight concrete	Timber Concrete	LCA Cradle to occupation	492 kg CO_2/m^2 569 kg CO_2/m^2	UK	Hacker et al. (2008)
Construction of office complex	Reinforced steel concrete	LCA	467 kg CO_2e/m^2	UK	Williams et al. (2012)
Timber-framed house with larch cladding	Larch and timber	LCA	405 kg CO_2/m^2	UK	Monahan (2013)

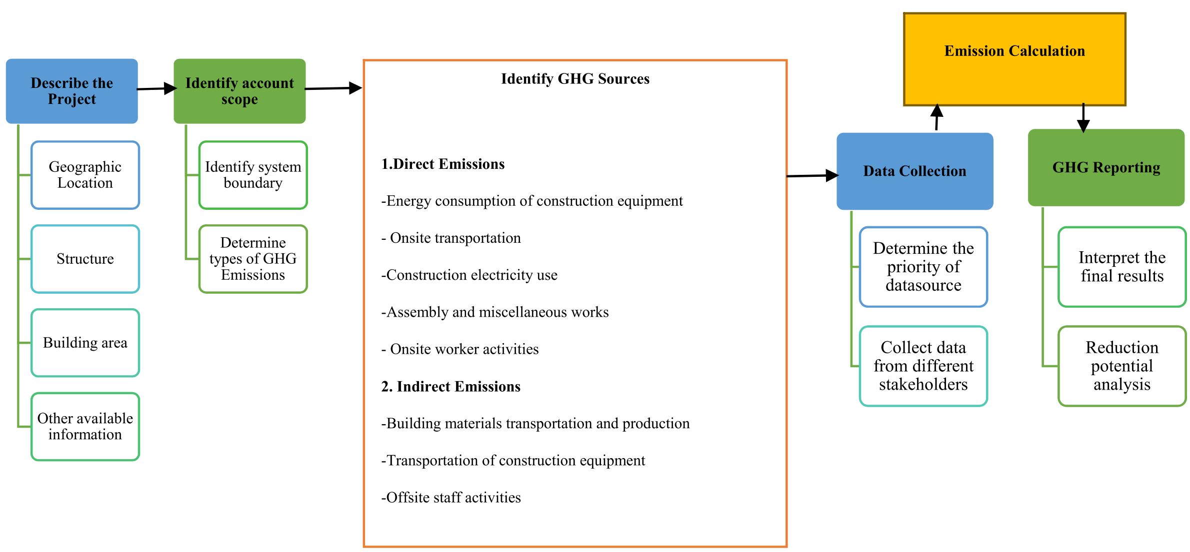

Fig. 2. Framework for carbon dioxide emissions calculation (Hong et al., 2015).

2013). Rice straws have been utilized as a means to construct hollow gypsum blocks, with the objective of producing exceptionally lightweight walls (Singh et al., 2022). Sugarcane bagasse ash (SCBA) is an efficient supplementary cementitious material (SCM) that has been studied for its slump, water absorption, porosity, microstructure, and mechanical properties in the production of green concretes (Agwa et al., 2022). Many studies by Bheel et al. and Rid et al. have called for the replacement of traditional cement with abundant agricultural and industrial waste, such as sugar cane bagasse, regenerated coal from waste, eggshell powder, and fly ash, in order to reduce dependency on non-renewable materials and minimize CO2 emissions (Bheel et al., 2022), (Rid et al., 2022). Krishna et al. (2020) looked into the best way to boost the ductility of concrete by incorporating coconut coir fibers, and they found that adding 1.5 percent of these fibers yielded the best results. They also found that using sisal fibers yielded the best results for boosting the strength of concrete. It has been found that lightweight concrete reinforced with natural fibers can significantly reduce heating and cooling costs (Asim et al., 2020). Lei Gu and Togay (Gu and Ozbakkaloglu, 2016) collated studies on the effects of plastic waste addition on concrete's characteristics and shape, as well as on plastic waste recycling methods. The use of recycled waste materials in pavement concrete was examined by Toghroli et al., in 2018 (Shariati et al., 2018). Recycled materials such as crushed glass, steel slag, steel fiber, tires, plastics, and asphalt have all been examined for their potential reuse. Polyethylene terephthalate (PET) and marble dust have been used to create composites for building (Krishna et al., 2020). According to another study by Haque and Islam (2021), PET plastic bricks were utilized to construct the Rohingya refugee camp. Another effort to lessen the world's reliance on fossil fuels, which contribute to greenhouse gas emissions (GHG), net-zero emissions buildings (NZEBs) have gained traction. In order to qualify as a NZEB, a building must have both a high level of energy efficiency and the ability to generate enough emission-free renewable energy to cover its energy needs for an established amount of time (Jaysawal et al., 2022). The American Institute of Architects (AIA) has put up a plan for a zero-emissions built environment by the year 2030. This paper is focused on the review of use of

Table 2
Analyzed research on the carbon dioxide emissions of building supplies.

Material used	Building Type	CO_2 Emissions	References
Steel	Highway tunnel	1.45 kg CO_2/m3	Hong et al. (2015)
	Residential complex	4.67 kg CO_2e/kg	Sreedhar et al. (2016)
	Highway construction	3.81 kg CO_2/kg	Monahan (2013)
	Timber frame house	0.43 kg CO_2e/kg	Arioglu et al. (2017)
	Highway tunnel	0.43 kg CO_2e/kg	Li et al. (2013)
Cement	Residential complex	0.759 kg CO_2/kg	Hong et al. (2015)
	Highway construction	0.8207 kg CO_2e/kg	Sreedhar et al. (2016)
		0.83 kg CO_2/kg	Jones and Hammond (2008)
		0.95 kg CO_2e/kg	Hammond and Jones (2011)
		1.165 kg CO_2e/kg	Arioglu et al. (2017)
Glass	Residential complex	1..09 kg CO_2/kg	Hong et al. (2015)
Timber	Residential Building	583 kg CO_2/m3	Hong et al. (2015)
Crushed Sand	Highway tunnel	0.005 kg CO_2e/kg	Arioglu et al. (2017)
Concrete	Timber frame house	0.1741 kg CO_2/kg	Monahan (2013)
	Residential complex	261 kg CO_2/m3	Hong et al. (2015)
Bitumen	Highway construction	0.426 kg CO_2e/kg	Sreedhar et al. (2016)
Gypsum	Timber frame house	0.30615 kg CO_2/kg	Monahan (2013)
Ceramics tile	Conventional building	0.6125 kg CO_2e/kg	Nyari (2015)
Wooden door	Conventional building	18.450 kg CO_2e/piece	Nyari (2015)
Fly ash	Highway tunnel	0.01 kg CO_2e/kg	Arioglu et al. (2017)
		0.01 kg CO_2e/ton	Li et al. (2013)
Superplasticizers	Highway tunnel	0.01 kg CO_2e/kg	Arioglu et al. (2017)
Lime	Highway construction	2.81 kg CO_2e/kg	Sreedhar et al. (2016)
Aggregates	Highway construction	0.0028 kg CO_2e/kg	Sreedhar et al. (2016)

economical green materials for construction sector including their energy consumption, carbon dioxide emissions, effect on physical and mental health and overall economy as discussed in the below sections.

2. Green building materials

Sustainable development encompasses a set of objectives that converge into three fundamental dimensions, referred to as the 3Ps: environmental, economic, and social pillars. The concept of sustainability, as outlined by Barbier (1987) (Barbier and Burgess, 2017), emphasizes the potential trade-offs that exist between various economic, environmental, and social objectives, each having their own positive or negative implications. Green building materials refer to materials that offer suitable performance and durability, requiring minimal maintenance, while simultaneously reducing the extraction of raw materials, pollution generated during manufacturing and use, and energy consumption. Additionally, these materials possess the highest potential for reuse or resource recovery (Rousseau).

2.1. Wood, bamboo, and straw bales

Wood possesses numerous advantages in comparison to contemporary industrial building materials such as concrete or steel, including lower upfront costs, greater ease of construction, a more aesthetically pleasing final product, and a longer useful life span provided it is kept dry and free of pests (Becker et al., 2011). Even though wood may bend under pressure without losing its integrity, it is at its strongest when squeezed vertically (Pacheco-Torgal and Labrincha, 2013). It has the much-desired ability to dampen noise and echoes, making it a valuable commodity. High-density rice straw elements coated in cement skins have been subjected to several tests (Thormark, 2006). After undergoing a 2-h fire testing procedure, it was seen that straw bales coated with cement plaster exhibited structural integrity, as neither the flame nor sufficiently high temperatures to ignite the interior straw of the bales were able to penetrate through the plaster to the opposite side. Scientists from the Fraunhofer Institute (Sharma et al., 2011), and Wilhelm Klauditz-Institute (WKI) in Braunschweig have created a new type of wood-insulating foam with exceptional properties. Instead of utilizing conventional high-impact plastic foams derived from petrochemical sources, this alternative wood foam can serve as a viable substitute. (Krieg et al., 2014). This resulting wood foam is highly insulating, resistant to moisture and pressure, and comparable in performance to standard plastic foams. Yuanyuan et al. introduced the use of translucent wood whose formation they mentioned in Biomacromolecules (Li et al., 2016). This procedure necessitates the elimination of lignin, a substance contained in the cell walls of wood. The glass in windows might be replaced with this translucent wood, which would make them cheaper and stronger. In addition to letting in natural light, the varying degrees of transparency in these windows also offer a measure of privacy.

2.2. Concrete

The inherent environmental sustainability of concrete as a construction medium is well recognized; nonetheless, environmental concerns mostly arise from the inclusion of cementitious elements in its composition. Approximately one metric ton of carbon dioxide (CO_2) is released into the atmosphere during the production of each metric ton of Portland cement. It is estimated that about 7% of all carbon dioxide emissions worldwide come from the cement sector (Malhotra, 1999). The utilization of fly ash as a significant pozzolan in conjunction with Portland cement is a convenient option. The utilization of fly ash in concrete mixtures offers several noteworthy advantages; including a low heat of hydration, improved strength and durability characteristics, cost-effectiveness, and a proportional decrease in CO_2 emissions (Iqbal and Jahangir, 2017). The utilization of recycled concrete as aggregate is a viable resolution to the challenges associated with the extraction of natural aggregates from quarries and the management of waste concrete (Bazant et al., 2000). To make new concrete mixes, crushed concrete aggregate can be recycled. The slump value of recycled concrete mixtures decreases dramatically because crumbled concrete absorbs more water than new coarse material. Recycled aggregates' surface irregularities also reduce their workability in mixes. Using a superplasticizer in a concrete mixture greatly reduces the aforementioned complications. If taken into account during the planning phase, a 13% drop in compressive strength caused by using 20% recycled crushed concrete could be acceptable (Batayneh et al., 2007). One of the green building materials that has been approved as being good for the environment is autoclaved aerated concrete (AAC). Porous, non-toxic, renewable, and recyclable, AAC can be used over and over again (Kalaipandian et al., 2018). The byproduct of thermal generating plants, fly ash, is used by AAC. While typical clay bricks require 8 kg of coal to produce one square foot of carpet area, AAC blocks only require 1 kg (Khandve and Rathi, 2015). To reduce energy consumption and pollution produced by regular cement, researchers from Mexico's Michoacan University of San Nicolás de Hidalgo developed a special cement that collects solar energy during the day and emits it at night (Han et al., 2017). Roads, highways, bike routes, and buildings could all benefit from the illumination provided by this phosphorescent cement. This revolutionary cement doesn't fade in the sun and has a 100-year lifespan. Martian concrete is 100% recyclable, uses no water, and is made from a combination of sulfur and raw minerals found in the Martian crust (Ghalandari et al., 2023). By incorporating bacteria into a base of sand, water, and trash, self-repairing concrete has been developed. This makes it possible for concrete to fix tiny cracks all by itself, without any help from humans (Jonkers and Schlangen, 2007). Designers have created a smog-eating cement that can withstand the effects of air pollutants over time. Cassar, 2014 conducted laboratory experiments and field research to conclude that a smog-eating cement façade on a five-story structure would increase construction costs by 15% but decrease air pollution by 20%. That means that air pollution could be cut in half if all the buildings in a major city were constructed with this material.

2.3. Recycled tires

Before people learned they could reuse old tires as construction materials, many discarded them instead of recycling them. Shredded, chipped, ground, or crumb rubber is utilized in cement composites and concrete for recycling (Bazant et al., 2000). A recycled tire crumb is a viable option for enhancing a variety of flooring types. By guaranteeing the necessary technical performance (such as moisture/vapor transfer, flammability, etc.), they can also be employed commercially on a wide scale as part of a roofing system (Malhotra, 1999). Crumbed rubbers are particularly useful as a waterproof membrane/damp-proof course to prevent water from seeping through the ground. According to Baricevic et al. each year, waste tyres yield over 250,000 tonnes of recycled tyre polymer fibre. Concrete has scattered reinforcement made of a variety of typical fibre kinds, including carbon, nylon, steel, polypropylene, glass, etc. As a result, recycled discarded tyres can provide a sustainable substitute for these common fibres, lowering building costs and detrimental consequences on the ecosystem (Baričević et al., 2018). As per Hambirao and Rakaraddi (2014), shredded rubber fibres can serve as a suitable replacement for reinforcement in deep foundations and raft foundations, resulting in increased unconfined compressive strength at optimal replacement ratios (Hambirao and Rakaraddi, 2014).

2.4. Ground granulated blast furnace slag

Ground Granulated Blast Furnace Slag (GGBFS) is a substance derived from the steel industry. It is a glassy granular material that is produced when molten blast furnace slag is rapidly cooled, typically through water immersion (Ainie Mat Dom et al., 2022). GGBFS exhibits cementitious qualities and possesses a specific gravity of 2.9. Experimental results have shown that the optimal degree of cement replacement ranges from approximately 50% to as high as 70% and 80% in certain cases. Similar to fly ash, this one also improves the mechanical and durability properties of concrete while decreasing the heat of hydration. Concrete made using GGBS cement has higher ultimate strength and a larger concentration of the strength-boosting calcium silicate hydrates (CSH) than concrete made with Portland cement. Hwan and Lin found that GGBFS has cementitious properties, making it a potential high-percentage substitute for cement (Hwang and Lin, 1986). Macphee raised concerns about the potential for corrosion of steel reinforcement in concrete containing slag cement due to changes in the pore solution. However, the study also suggests that the altered pore structure of slag concrete may actually help to reduce corrosion by increasing the resistivity of the concrete (Macphee and Cao, 1993). Hussain (Hussain et al., 1995) reported that increasing the C3A content in concrete from 2% to 14% led to a 2.43-fold and 2.45-fold increase in chloride binding capacity and reinforcement corrosion initiation time, respectively. This suggests that C3A plays a crucial role in enhancing concrete's resistance to chloride ingress and corrosion protection of embedded steel. Yeau and Kim (2005) investigated the impact of GGBFS replacement on the corrosion resistance of two Portland cement types (I and V) through various tests like permeability, diffusivity, and corrosion area analysis. They concluded that Type I cement concrete exhibited lower permeability and protected the steel better compared to Type V. Increasing GGBFS replacement in Type I further enhanced its corrosion resistance against chloride ingress and potential steel damage.

2.5. Glass bricks

This invention, as documented in the Glass Structures & Engineering journal (Oikonomopoulou et al., 2018), endeavors to amalgamate the conventional architectural style of the city with the global architecture prevalent in major urban centers. Within the structure, a deliberate arrangement of glass bricks and traditional clay bricks is observed in a slow alternation as one ascends, creating an illusion of the building levitating over the roadway (Oikonomopoulou et al., 2018). Advanced laser technology was employed to cut bricks, while laboratory-grade UV lamps were utilized to facilitate the curing process of the adhesive employed in compacting the bricks. Milk was employed due to its diminished transparency, rendering it a suitable liquid for the creation of reflected surfaces and the first leveling of bricklayers. The glass bricks exhibit exceptional durability, surpassing that of concrete, and possess the ability to endure the impact of a vehicular collision. Indeed, the glass construction was found to possess superior strength compared to concrete, as demonstrated by the strength tests conducted by the team from Delft University of Technology. One of the primary benefits associated with glass is its total recyclability. As an illustration, during the installation process, several flawed bricks were subjected to melting and subsequent remolding.

2.6. Eco-friendly tiles

An environmentally friendly tile that can be used as an alternative to traditional flooring because it is made using less energy. When compared to regular tile, the price is far lower. They are simple to install and come in a wide variety of designs to meet the needs of any customer. This tile enhances the efficiency of the interior environment. When compared to traditional tiles, eco-friendly options are more affordable since they are produced on-site, eliminating the need for costly transportation (Bakhoum et al., 2015).

2.7. Biocomposites

Biodegradable materials are manufactured from renewable resources that break down when exposed to microbes, leaving behind only innocuous byproducts (Energy). Natural fibers are used in bio-composite reinforcement, whereas biopolymers are used in bio-composite matrices (Energy) (Wong et al., 2002). In comparison to the polymeric matrix, natural fibers are more rigid and robust. PHB is an organic and biodegradable polymer that has many benefits over petrochemical plastics, such as biodegradability, the use of a low-cost renewable carbon source (sugar or corn starch), the use of less energy during production, and the release of fewer greenhouse gas emissions throughout its life cycle. Natural fibers (NFs) are a viable choice as a reinforcing element for bio-composites owing to their relatively high specific strength and stiffness. Kenaf is a bast fiber with huge potential as a reinforcing fiber in composites and a priority for choosing green materials, as it absorbs more carbon dioxide than any other plant (one ton of kenaf absorbs 1.5 tons of ambient CO_2) (Akil et al., 2011). The most significant studies examining the characteristics and roles of bast fibers in composites and bio-composites are shown in Table 3, proving that these entirely bio-based materials can be used safely and effectively in building.

Table 3
Biocomposites reinforced with bast fibres suggested for use in construction.

Type of Biocomposite	Findings	References
Cellulose acetate/hemp fiber composites	Investigated the foundation's thermal, morphological, and rheological properties	Joon soon et al. (2005)
Cellulose acetate/hemp fiber composites	Foundational physics and thermomechanics were studied in detail.	Mohanty et al. (2004)
Hemp reinforced biocomposites	Wood and other woody materials have comparable qualities in the construction sector	(Joon soon et al., 2005)– (Christian and Billington, 2009)
Hemp cellulose acetate/Polyhydroxybutyrate	Has mechanical properties that are analogous to those of structural wood	Christian and Billington (2009)
Combining PHB and its several co-polymers with hemp fiber	Exhibit desirable mechanical properties, suggesting potential for application in building.	(Keller, 2003), (Taha and Ziegmann, 2006)
Combining PHB and its co-polymers with flax fiber	They highlighted the versatility of biocomposites and their prospective uses in temporary construction, such as scaffolding, formwork, flooring, and walls.	Bodros et al. (2007)
Polyhydroxybutyrate (PHB) and its several co-polymers with jute fiber		Khan et al. (1999)

2.8. Insulated concrete forms (ICF)

Insulated concrete forms (ICFs) consist of a construction system wherein layers of insulation materials are utilized to encase concrete walls. The aforementioned technology exhibits a high level of energy efficiency while concurrently enhancing the structural integrity of the walls. This is recognized for its robustness and extended lifespan. The material is mostly utilized as a construction component in low-rise structures intended for residential and commercial purposes. U.S. Housing and Urban Development Carried Out a Study in 2001 revealed that the cost of modular construction is approximately 3–5% more compared to traditional building methods (Bazant et al., 2000). Apart from cost considerations, there are numerous performance advantages such as fire resistance and energy efficiency.

3. Emissions of green building materials concerning conventional building materials

In this section, several scholarly articles were examined. The selection of these articles was based on their ability to effectively demonstrate the advantages associated with the construction of sustainable buildings. These studies highlight the potential and advantages of adopting a sustainability-focused strategy for building in a fast-emerging country seeing significant growth in its construction industry. Both publications examined prototypes constructed using environmentally friendly building materials as well as conventional materials. In their study, Bakhoum et al. (2015)utilized a Sustainable Decision Support System (SDSS) to conduct a comparative analysis of different structural building materials. On the other hand, Lalit Srikar (Srikar et al., 2022) employed Auto CAD 2021 and SketchUp software for rendering the design of building models, while STACK software was utilized for quantity and cost calculation purposes. Bakhoum et al. examined the sustainability evaluation of materials throughout three distinct periods of their life cycle: production, construction, and demolition. Life cycle inventory (LCI) information for building materials in both green and traditional structural systems is displayed in Fig. 3. The findings from Bakhoum et al.'s paper (SDSS) indicate that the Eco-friendly system outperformed the traditional method throughout all three phases. The data presented in Fig. 4 demonstrates that the environmentally conscious system showcased 64.7%, 64.4%, and 84.6% sustainability rates for the manufacturing, construction, and demolition stages, respectively. In contrast, the conventional system achieved sustainability percentages of 54%, 63.9%, and 75.7% for the same phases.

The scores for sustainable elements in the two studied systems are depicted in Fig. 5. The Ecofriendly system exhibited higher sustainable scores across all sustainable factors in comparison to the conventional system. This can be attributed to the significantly greater quantities of materials utilized in the latter system, with conventional materials

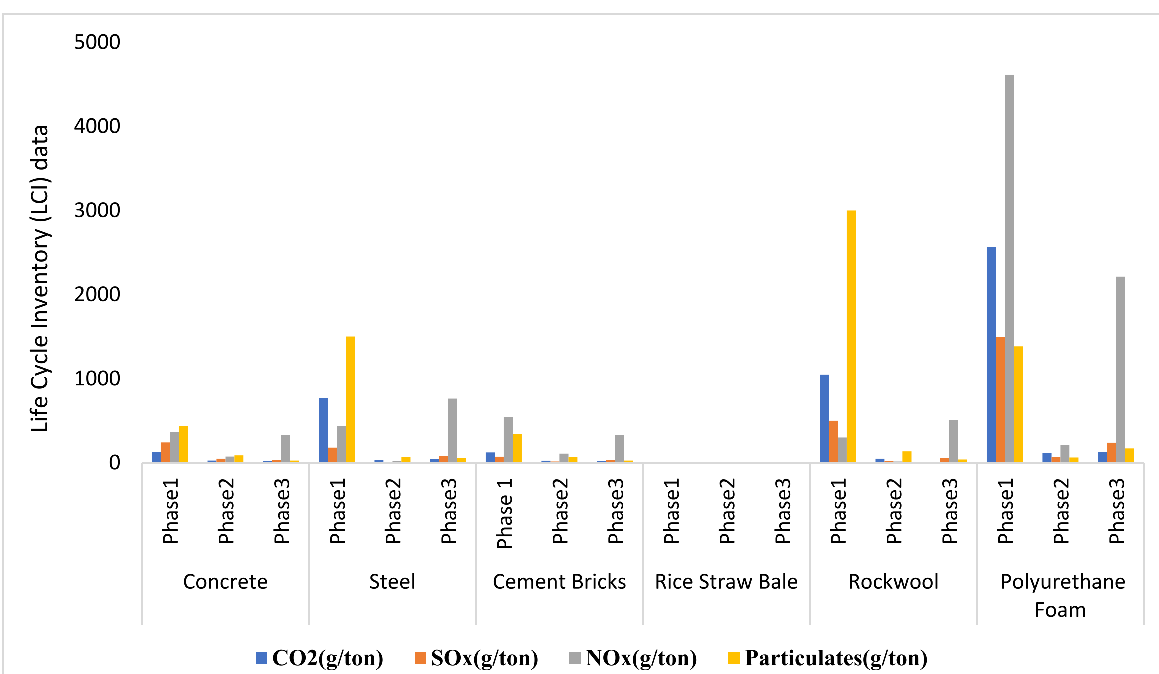

Fig. 3. LCI statistics for recycled materials.

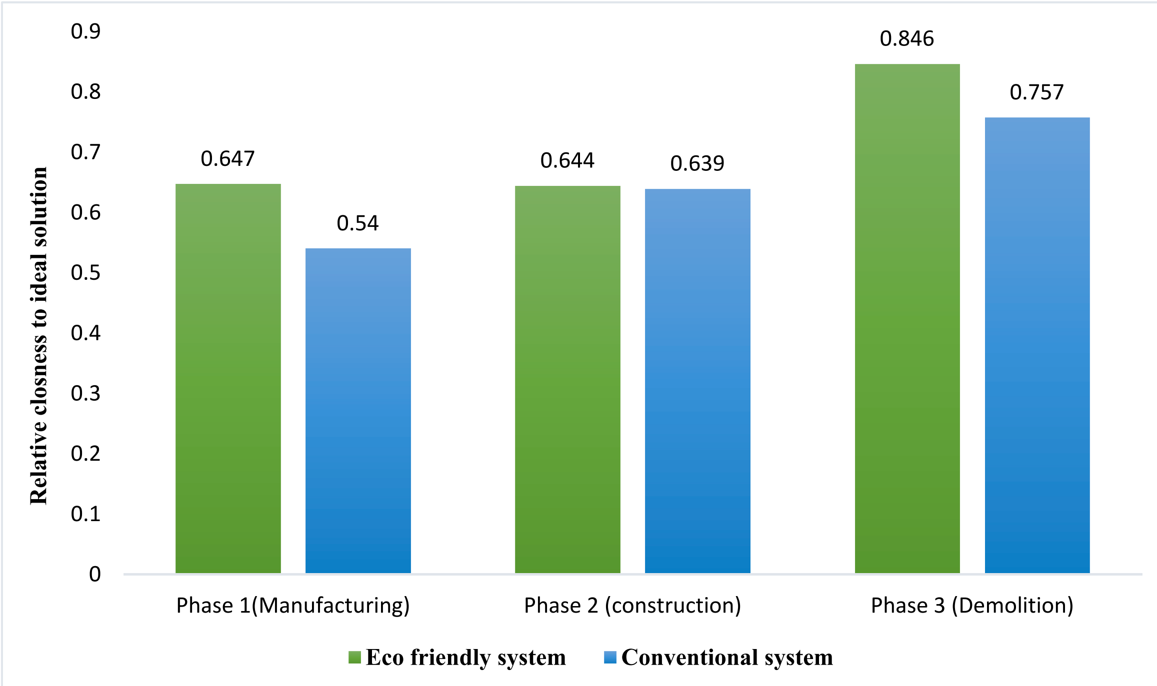

Fig. 4. Sustainable evaluation of Eco-friendly and conventional systems through the different phases.

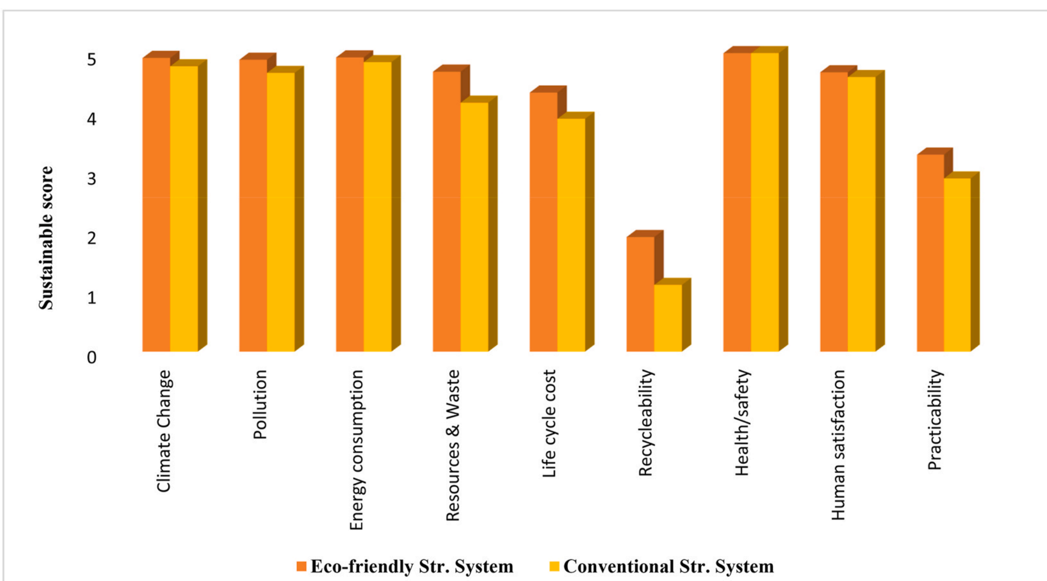

Fig. 5. Sustainable factors scores for Eco-friendly and conventional systems (Phase I.

exceeding eco-friendly materials by more than 2.5 times. The Eco-friendly method had a greater resources/waste factor score compared to the traditional system, with a difference of 9% in phase II and 35% in phase III, representing the highest disparity observed between these phases. However, the Eco-friendly system had a drop of 6% in its reuse factor score during phase III, when compared to the traditional method. According to the results obtained from the Sustainable Development Sustainability System (SDSS), the eco-friendly system demonstrated a higher sustainability rank of 67% compared to the conventional system, which achieved a sustainability rank of 56%. This information is visually shown in Fig. 6.

Lalit Srikar in his paper offered a comparison table of embodied energy as well as a comparison chart of power usage. According to the data presented in Fig. 7, the calculation of embodied carbon indicates that the sustainable construction model has the potential to achieve annual savings of up to 36.5%. This measure will significantly ease the environmental consequences associated with the construction of the structure. The utilization of locally sourced materials can contribute to the enhancement of the local economy. The incorporation of several sustainable features such as a solar power system, rainwater harvesting, and the use of alternative building materials significantly reduces the electricity consumption of the sustainable building. The annual savings percentage was determined to be 44.4% as visible from Fig. 8.

Yu-Hsiang et al. (Cheng et al., 2015) compared and tested eight building materials (ceiling, cabinetry, flooring): calcium silicate, mineral fiber, gypsum, and wood, each in both conventional and "green"

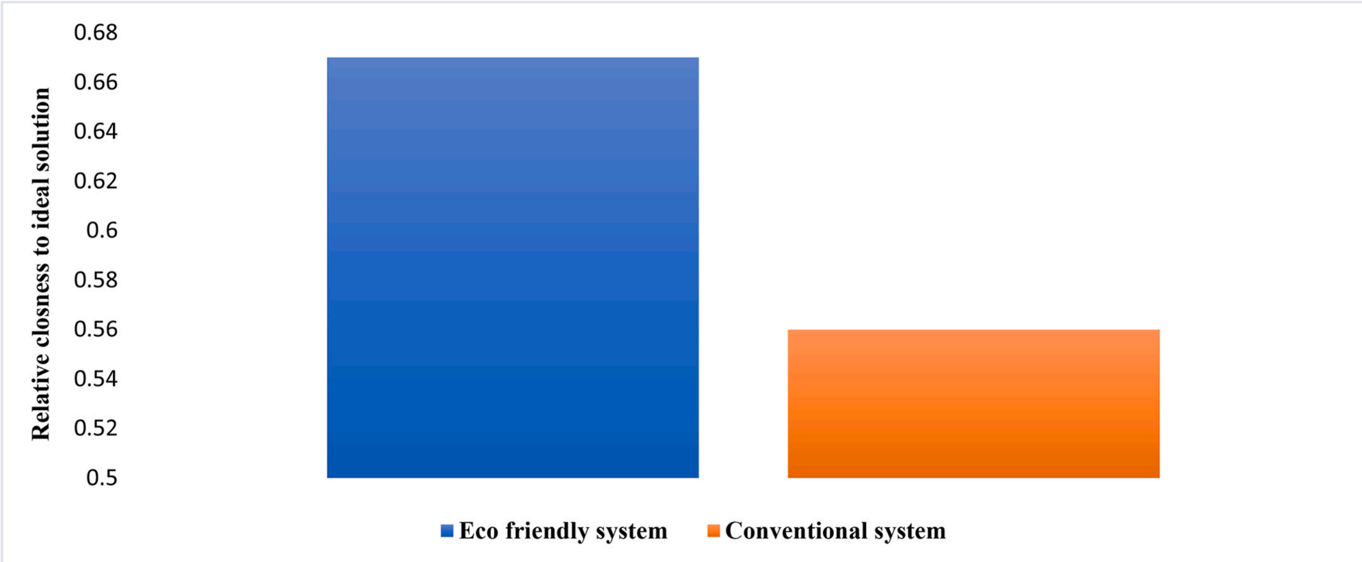

Fig. 6. Overall sustainable evaluation for eco-friendly and conventional systems.

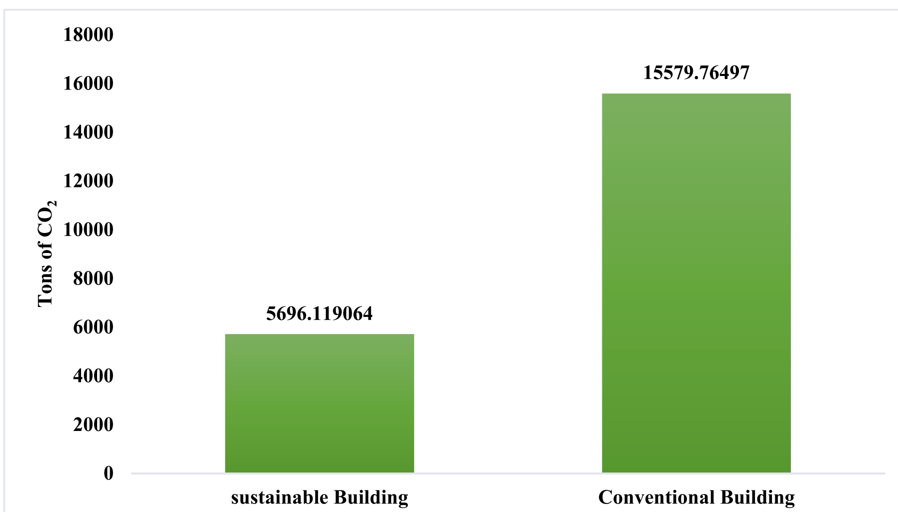

Fig. 7. Embodied energy comparison chart.

versions. Green materials exhibited lower primary emissions of carbonyls and BTEX, along with reduced molar yields and secondary carbonyl emissions, especially for wood flooring and gypsum board. Both green and conventional pairs reacted with ozone. While conventional materials are typically cheaper, green options boast lower VOC emissions. Fleecy and porous materials displayed higher ozone reactivity compared to smooth and nonporous surfaces. The observed secondary emissions suggest moderate indoor ozone concentrations can significantly increase carbonyl levels, particularly formaldehyde and acetaldehyde, potentially raising chronic health concerns for some materials (CSB, WF, GCSB, GGB). This emphasizes the need for improved building material testing protocols to encompass heterogeneous ozone-material interactions.

Su et al. (2021) researched green building technologies in Dalian and found that high-performance building envelopes (outer layers) were the most cost-effective and impactful, with 73% adoption. While green materials are good for the environment, they must also meet basic human needs for wider green building acceptance. The current focus for carbon emission reduction is improving the effectiveness of green materials so they can effectively replace traditional materials with high energy consumption and emissions in construction. Zhao et al.'s (Zhao et al., 2020) work highlights the immense potential of energy theory in guiding the development of sustainable construction materials. By prioritizing green concrete and continuously improving its efficiency, we can pave the way for a future where our built environment thrives in harmony with our planet. Wu et al. (2017) in their research delves into the carbon emissions of green buildings, pitting them against their non-green counterparts. Using a sophisticated life cycle assessment approach fueled by the BELES(Building Environmental Load Evaluation System) platform and data from 26 real buildings, the study yields revealing findings: Cross-Laminated Timber (CLT) systems reign supreme in minimizing Life Cycle Carbon Emissions (LCCO2) for both residential and commercial buildings, Green commercial buildings shine brighter, boasting a 32% reduction in LCCO2 compared to their non-green counterparts. Residential green buildings also contribute, but with a 10% drop. This suggests more potent carbon-reduction strategies within the commercial green realm.

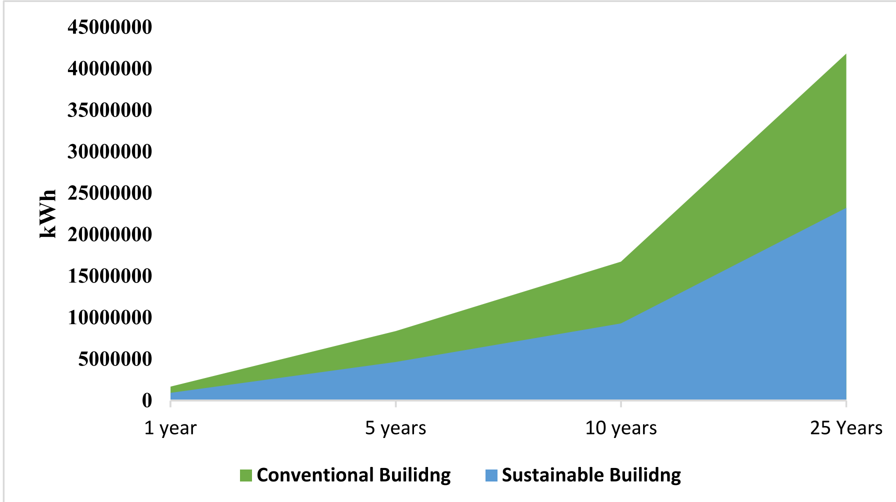

Fig. 8. Operational Life Electricity usage comparison chart.

4. Green building materials and health

The impact of construction materials on human health is an unavoidable factor. However, the shift from traditional building materials to environmentally friendly materials has been implemented in order to mitigate the overall impact on human health in both indoor and outdoor settings. The utilization of green building materials (GBMs) including non-toxic, natural, and organic components has promise in addressing the decline in indoor air quality (IAQ) and minimizing the overall negative effects on human health (Persily, 2015). Previous studies have primarily concentrated on examining the effects of incorporating environmentally friendly design, materials, and building methods on the well-being and efficiency of individuals who ultimately utilize a given facility. The World Health Organization (WHO) has developed a comprehensive guideline to enhance indoor air quality (IAQ) standards. Based on the established guidelines, indoor pollution originates from four primary sources. The available resources include building materials, household electrical appliances, heaters, and soot generated by culinary activities. According to LIQUN and Yanqun (Liqun and Yanqun, 2011), buildings and ornamental materials were identified as the primary contributors to indoor pollution. The primary detrimental effects arising from these sources are radioactive and chemical contamination. Decorative materials and elements, such as brick walls, stones, and concrete, have the potential to emit radioactive substances. Chemical contamination emanates from various sources, including the textile, paint, and glue industries. Both of these factors have the potential to contribute to the development of cancer over an extended period, posing a significant risk to the health of individuals residing in the affected environment, particularly youngsters. Indoor chemical pollutants and their negative effects are given in Table 4.

As per the United States Green Building Council (2006), "green building" refers to a set of practices that aim to lessen adverse effects on the environment and the economy. Surveying the building's inhabitants is the quickest and easiest technique to gauge IEQ and gauge their level of contentment within it (Ramsay et al., 1990). The Center for the Built Environment at the University of California, Berkeley developed a tool called "cognitive interviewing" to test the comprehension and accuracy of replies from building occupants in several research (Huizenga et al., 2003). Across the board, research shows that people who work in LEED-certified buildings are happier with the conditions in those facilities. Unlike Altomonte et al. (Altomonte, 2013), Huizenga et al. (2003), Lee et al. (Zakeri et al., 2022), and Abbaszadeh et al. (2006) all discovered that inhabitants of LEED buildings reported greater satisfaction with thermal comfort. When comparing the lighting and acoustics of various building styles, neither Huizenga nor Abbaszadeh identified any statistically significant differences. The literature review on the topic of green buildings and health is summarized in Table 5. The literature on the correlation between green construction materials and IEQ metrics is summarized in Table 6.

5. Impact of green building materials on economics

Nalewaik and Venters (2010) found that when green sustainable buildings first entered the public consciousness, there was a consensus that they were more expensive to construct. According to Aneta (Mesjasz-Lech and Włodarczyk, 2022), many property owners are not reaping the financial benefits of green building construction or retrofits because they are not employing energy-efficient technologies or sustainable operational approaches. According to Buys and Hurbissoon (Buys and Hurbissoon), it's important to take into account both direct and indirect

Table 4
Major indoor air pollutants and their negative effects.

Pollutants	Negative Effects	References
Benzene	Bone marrow damage, thrombopenia, leukopenia, anemia	World Health Organization (2010)
Toluene	Extremely toxic to the liver, blood, nerves, and mental health.	Manisalidis et al. (2020)
Xylene	Harmful to the nervous system	Kandyala et al. (2010)
Ethylbenzene	Extremely harmful to the nerves.	Masekameni et al. (2018)
Styrene	Eye irritation, central nervous system atrophy, and acute toxicity are all possible side effects.	Werder et al. (2018)
HCHO	Exposure to this substance can cause a variety of health problems, including skin diseases, rhinitis, emotional instability, memory loss, nerve damage, cancer, eye, nose, and throat irritation, cough, diarrhea, vertigo, and nausea.	Liu et al. (2023)

Table 5
An analysis of the literature on the health benefits of green buildings.

Building Type	Results	References
Offices – 3 green – 45 conventional	↑ Air quality ↑ Cleanliness	Huizenga et al. (2003)
Offices – 21 green – 160 conventional	↑Air quality ↑ Cleanliness ↑ Thermal comfort	Abbaszadeh et al. (2006)
Offices – 15 green – 200 conventional	↑Air quality ↑ Cleanliness ↑ Thermal comfort ↓ Lighting ↓ Acoustics	Altomonte (2013)
Offices – 65 green – 79 conventional	↑Air quality ↑ Cleanliness ↓ Thermal comfort ↓ Lighting ↓ Acoustics	Altomonte (2013)
University – 2 green – 1 conventional	↑Air quality ↑ Ventilation	Hedge et al. (2014)
Offices – 2 green – 1 conventional	↑ Self-reported well-being ↓ Absenteeism ↓ Asthma and allergy symptoms	Singh et al. (2010)
Offices – 3 green – 2 conventional	↑Air quality ↑ Lighting ↑ Thermal comfort ↑ Acoustics	han-hsi Liang et al. (2014)
Factories – 1 green – 1 conventional	↓ Thermal comfort ↓ ventilation	Ravindu et al. (2015)
Offices – 12 green – 12 conventional	↑Air quality ↓ Acoustics	Newsham et al. (2013)
Public housing – **1 green rehabilitation**	↑ Self-reported well-being ↑ Cleanliness ↓ Allergens	Jacobs et al. (2014)

benefits when calculating the true cost of going green. Lower operating expenses, higher capital values, and more rental revenue are all advantages of a green building. Eco-friendly structures improve not just the health of the building's inhabitants, but also their productivity, marketability, and public perception as a whole. In Table 7 we see the monetary gains from using environmentally friendly construction methods and materials.

6. Conclusion

This paper comprehensively explores the impact of green materials on sustainable development in the context of building construction. By citing a multitude of studies and drawing upon diverse sources, the paper underscores the myriad advantages associated with this transformative construction methodology. Key discussion points include the diverse benefits of green materials, ranging from a notable decrease in environmental impact to improvements in the physical and mental health of occupants. The paper emphasizes the crucial role of green materials as instruments for sustainability, aligning strongly with international objectives related to climate change, resource preservation, and resilient communities. Furthermore, it delves into the positive effects on carbon footprint reduction and energy efficiency, ultimately leading to significant cost reductions over a building's lifespan. Addressing the crucial aspect of indoor air quality, the paper highlights how green materials contribute to a healthier living environment by reducing allergens and harmful substances. Despite their potential advantages, the paper acknowledges challenges in the implementation of green materials, such as early financial impediments and the need for specialized expertise in the building sector. The collaborative nature of solutions is stressed, advocating for close partnerships among policymakers, industry stakeholders, and researchers to advance, make available, and cost-effectively integrate green materials and technologies. Ultimately, this paper concludes that green materials are not merely optional but fundamental components of any sustainable development strategy. The presented findings and insights serve as a robust foundation for informed decision-making regarding the use of environmentally friendly materials in construction methods.

Table 6
Literature on the relationship between green buildings and IEQ parameters.

IEQ parameter	References	Remarks
Indoor Air Quality (IAQ)	No Title (2000)	Sick building syndrome (SBS) and numerous types of respiratory allergies were discussed, and their causes were determined to be NOx, CO, and RSPs. Exposure to ozone may cause hypertension and impaired cognitive growth in children, as well as other health concerns.
	Wargocki et al. (2000)	That low quality of indoor air will lead to rising SBS and a drop in residents' performance
	Garland et al. (2013)	56% who work in LEED-certified buildings are happy with the air quality.
Temperature	Seppänen et al. (2002)	If the temperature in an indoor environment rises above 25°C, productivity will decrease.
	Fog and Version (2001)	An essential role in the growth of molds inside the buildings
	Paul and Taylor (2014)	Buildings with thermal control devices have workers who are 11% more productive than those in traditional buildings.
Humidity	Ries et al. (2006)	High humidity can cause health problems for residents and increase the risk of respiratory illnesses
Ventilation	Yu and Kim (2011)	Pollutants in the outside air must be removed before it enters the building. A building with good ventilation will have better air quality, happier people because of the pleasant temperature, and fewer negative consequences on their health and happiness. Most green building rating methods prioritize ventilation.
Lighting	Hwang and Kim (2011)	Poor lighting can cause a variety of health problems, including allergic reactions, dry eyes, headaches, and eye fatigue.
	Romm and Browning (1998)	employees' output and efficiency gain noticeably as a result of green workplaces' better lighting.
Acoustics	Evans and Stecker (2004)	Sound pollution negatively affects occupants' performance and productivity
Occupant Health and Wellbeing	(Goe et al., 2004)– (Moorman et al., 2007)	Mold and dust are the most common indoor allergens because they thrive in warm and humid conditions, where they produce liquids and gases that can trigger allergic reactions in people.
	Fletcher et al. (2006)	A humid environment, visible mold and mold odor, high temperatures, and indoor pollutants can create conditions where allergens and irritants can thrive.
	Heschong et al. (2003)	Employees who have access to natural views and improved ventilation and temperature conditions in their work environment are more likely to be productive and perform better.
	Huizenga et al. (2003)	The quality of the indoor environment, such as the air quality, lighting, and temperature, can have a significant impact on people's psychological health.
	Seppänen et al. (2002)	Increases in indoor environmental quality in the United States saved an estimated \$6–14 billion per year, with an additional \$1-\$4 billion gained from fewer incidences of asthma and allergies among workers. These financial gains can be supplemented by a decrease in the death rate from radon-related lung cancer and carbon monoxide poisoning.

Table 7
Financial benefits of using green materials in buildings.

Financial Benefit	References	Remarks
Reduction in Electricity consumption	(Milne)	Typical energy efficiency gain for a green building is between 25 and 30 percent.
	(Malarthamil)	Energy savings of up to 35% can be achieved with the implementation of full heat recovery technology in a central air conditioning system.
Less use of water	Watson (2009)	Buildings that have been designed with sustainability in mind use significantly less freshwater, which has a favorable impact on a business's bottom line.
Productivity	Solutions (2012)	A green building's healthy indoor atmosphere translates to fewer sick days, decreased absenteeism, and higher retention rates among employees.
	Ries et al. (2006)	There was a roughly 25% increase in output, a statistically significant shift in absenteeism, and a 30% drop in energy use per square foot.
Ability to attract and retain corporate and government tenants	(Mattson-Teig)	Tenants may be more interested in renting a building if it has been given a "green title."
	(Milne)	Property owners and managers reap significant benefits from green buildings due to the low vacancy rates and high tenant retention rates.
Reduce liability and risk	(Milne)	By constructing sustainably and implementing eco-friendly practises now, property owners can protect themselves against the high cost of retrofits and future obsolescence.
	(Yudelson)	Green building certification can help to protect against future claims about indoor air quality by having a third party verify that the building meets certain standards, beyond just following construction codes..
Higher return on investments	Solutions (2012)	There is potential for significant energy savings in new and existing buildings in the US, up to 9.9% and 19.2%, respectively.
Marketability	(International and Solutions)	A project's marketing budget benefits from free publicity, while the project benefits from increased occupancy and higher rental prices as a result of increased demand from tenants.
	Leaman and Bordass (2007)	Employees and visitors to green buildings like the enhanced working conditions, while business owners appreciate the positive financial impact.

CRediT authorship contribution statement

Tagamud Tazmeen: Conceptualization, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Fasil Qayoom Mir:** Conceptualization, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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